

WASHINGTON, IA, USA

WMO: 725454

Lat: 41.276N

Lon: 91.673W

Elev: 230

StdP: 98.59

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04903

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -19.8	(c) -17.0	(d) -23.7	(e) 0.5	(f) -19.5	(g) -21.2	(h) 0.6	(i) -15.7	(j) 13.6	(k) -8.2	(l) 12.1	(m) -5.6	(n) 4.1	(o) 290	(p) 0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	33.8	25.4	32.5	25.0	31.1	24.1	27.5	31.5	26.3	30.5	25.3	29.5	4.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	22.4	30.2	25.1	20.7	29.1	23.9	19.3	28.1	88.7	31.8	83.5	30.5	78.5	29.6	32.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.6	(b) 10.3	(c) 8.9	(e) -24.4	(f) 36.9	(g) 3.8	(h) 4.7	(i) -27.1	(j) 40.3	(k) -29.4	(l) 43.1	(m) -31.5	(n) 45.8	(o) -34.3	(p) 49.2		
(a) 11.6	(b) 10.3	(c) 8.9	(e) -23.3	(f) 28.7	(g) 4.8	(h) 1.5	(i) -26.8	(j) 29.8	(k) -29.6	(l) 30.7	(m) -32.3	(n) 31.5	(o) -35.8	(p) 32.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-4.1	-2.4	4.3	11.1	17.4	22.5	24.0	23.0	18.5	12.0	4.5	-1.9
	DBStd	11.34	6.91	6.75	6.52	5.41	4.71	3.45	3.35	3.07	4.42	5.50	5.86	6.32
	HDD10.0	1633	438	349	199	49	2	0	0	0	2	41	183	369
	HDD18.3	3325	696	581	437	222	77	6	2	3	51	209	416	626
	CDD10.0	1925	0	2	22	83	231	374	434	402	256	103	17	2
	CDD18.3	577	0	0	2	7	46	131	178	147	54	13	1	0
	CDH23.3	5634	0	0	13	102	466	1225	1817	1299	580	129	4	0
	CDH26.7	2068	0	0	1	23	146	458	749	456	201	33	0	0
Wind	WSAvg	4.1	4.8	4.8	5.0	5.2	4.5	3.7	2.9	2.6	3.0	4.0	4.6	4.5
Precipitation	PrecAvg	962	35	30	64	91	121	116	117	102	105	76	60	44
	PrecMax	1539	106	104	150	228	292	285	325	300	327	203	182	160
	PrecMin	531	0	0	8	16	30	10	13	8	1	3	1	7
	PrecStd	217	25	22	37	53	71	64	69	61	71	50	43	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	17.6	25.2	30.0	32.8	34.2	37.0	34.1	33.8	30.2	23.0	17.2
		MCWB	8.9	11.5	16.2	20.0	22.4	24.9	25.9	26.1	23.9	21.9	15.2	14.2
	2%	DB	10.2	12.9	21.2	27.0	30.1	32.7	34.0	32.5	31.3	27.2	19.8	12.5
		MCWB	7.5	8.6	14.4	16.9	20.7	24.6	26.2	25.8	23.1	18.8	13.8	9.9
	5%	DB	7.2	9.8	17.5	23.7	27.8	31.3	32.6	31.1	28.9	23.9	17.0	9.0
		MCWB	4.8	7.2	12.3	15.5	19.3	23.5	25.8	24.9	22.0	17.9	12.2	7.0
	10%	DB	4.1	6.3	14.1	21.1	26.0	29.1	31.0	29.1	27.1	21.2	13.7	6.3
		MCWB	2.3	4.0	9.5	13.8	18.3	22.6	24.9	23.8	20.2	15.9	9.5	4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	13.0	17.9	20.6	24.8	28.2	29.3	28.3	25.8	23.0	17.3	15.1
		MCDB	12.0	17.5	23.6	26.7	29.3	31.8	32.9	33.0	31.0	28.2	20.7	17.2
	2%	WB	7.2	9.6	15.5	18.2	22.9	26.2	28.0	26.8	24.5	20.9	14.8	10.5
		MCDB	9.9	13.1	19.6	24.5	28.1	30.3	31.9	31.0	29.5	24.1	18.5	11.8
	5%	WB	4.8	6.8	13.1	16.5	21.5	25.0	26.8	25.7	22.9	18.9	12.5	7.1
		MCDB	7.2	9.5	17.3	22.0	26.2	29.5	30.8	29.7	27.3	22.3	16.2	8.6
	10%	WB	2.7	4.1	10.1	14.9	19.9	23.8	25.6	24.6	21.6	16.9	9.9	4.3
		MCDB	4.3	6.5	13.9	19.9	24.0	28.1	29.8	28.4	25.6	20.5	13.2	6.2
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.0	10.4	12.1	11.6	10.9	10.9	10.9	12.9	11.7	10.2	8.2
		MCDBR	11.9	13.6	15.0	16.7	13.9	12.6	11.9	12.2	14.9	14.8	14.3	11.3
		MCWBR	9.3	9.7	9.2	9.1	6.8	6.1	6.2	7.0	7.3	8.0	9.5	8.9
	5% WB	MCDBR	11.5	12.6	13.5	14.0	12.2	11.0	10.8	10.7	12.4	11.8	13.1	10.6
	MCWBR	9.2	9.5	8.9	8.9	7.2	6.6	7.0	6.9	7.0	7.5	9.6	9.1	
Clear-Sky Solar Irradiance	taub	0.289	0.304	0.338	0.383	0.423	0.422	0.440	0.422	0.389	0.342	0.301	0.290	
	taud	2.427	2.390	2.385	2.269	2.223	2.279	2.211	2.274	2.322	2.454	2.501	2.443	
	Ebn at Noon	870	908	912	887	856	855	835	839	844	853	851	836	
	Edh at Noon	78	95	107	130	139	132	140	128	114	89	73	71	
All-Sky Solar Radiation	RadAvg	1.84	2.82	3.76	4.73	5.55	6.18	6.19	5.44	4.63	3.07	2.07	1.53	
	RadStd	0.21	0.23	0.31	0.37	0.40	0.61	0.43	0.32	0.36	0.34	0.17	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page